

Bulan 3: Advanced Django Features

Minggu 1: Class-Based Views & Generic Views

Hari 1: Function vs Class-Based Views

Project: ViewLab

- Deskripsi: Comparison tool showing FBV vs CBV implementations
- Fitur: Side-by-side code, performance comparison, use case guide
- Keunikan: Interactive converter FBV ↔ CBV

Hari 2: TemplateView & ListView

Project: ProductCatalog

- Deskripsi: E-commerce product listing dengan filtering dan pagination
- Fitur: Category browse, search, sort, infinite scroll
- Keunikan: Smart filters yang adapt based on available products

Hari 3: DetailView & CreateView

Project: RecipeShare

- Deskripsi: Recipe sharing platform dengan step-by-step guide
- Fitur: Recipe details, ingredient calculator, cooking timer
- Keunikan: Voice-guided cooking mode

Hari 4: UpdateView & DeleteView

Project: TaskMaster

- Deskripsi: Task management dengan drag-drop prioritization
- Fitur: CRUD tasks, subtasks, deadlines, categories
- Keunikan: Undo delete dengan soft-delete pattern

Hari 5: FormView & AJAX Integration

Project: ContactFlow

- Deskripsi: Contact form dengan real-time validation dan file attach

- Fitur: Multi-step form, captcha, spam filter
- Keunikan: Smart suggestions berdasarkan input type

Hari 6: Mixins & Multiple Inheritance

Project: MixinToolkit

- Deskripsi: Reusable mixin library untuk common patterns
- Fitur: LoginRequired, PermissionRequired, PaginationMixin
- Keunikan: Auto-generate mixin combinations

Hari 7: Custom Generic Views

Project: CalendarView

- Deskripsi: Calendar interface dengan event management
- Fitur: Month/week/day views, recurring events, export iCal
- Keunikan: Natural language event creation

Minggu 2: Django REST Framework Basics

Hari 8: REST API Concepts & Setup

Project: APIStarter

- Deskripsi: REST API boilerplate dengan best practices
- Fitur: Serializers, versioning, documentation
- Keunikan: Auto-generate Postman collection

Hari 9: Serializers

Project: BookAPI

- Deskripsi: Library API dengan nested serialization
- Fitur: Books, authors, genres dengan relationships
- Keunikan: Smart serializer yang adapt output based on user role

Hari 10: API Views & ViewSets

Project: MovieDB API

- Deskripsi: Movie database API dengan TMDB-style endpoints

- Fitur: Search, filters, ratings, reviews
- Keunikan: Recommendation engine integration

Hari 11: Authentication & Permissions

Project: SecureAPI

- Deskripsi: API dengan JWT authentication dan rate limiting
- Fitur: Token refresh, blacklist, scope-based permissions
- Keunikan: API key management dashboard

Hari 12: Pagination & Filtering

Project: DataAPI

- Deskripsi: Large dataset API dengan advanced querying
- Fitur: Cursor pagination, field filtering, search
- Keunikan: GraphQL-style field selection

Hari 13: File Upload API

Project: MediaAPI

- Deskripsi: Media upload API dengan chunked upload support
- Fitur: Progress tracking, resumable uploads, CDN integration
- Keunikan: Client-side compression before upload

Hari 14: API Documentation

Project: SwaggerDocs

- Deskripsi: Auto-generated API documentation dengan Swagger UI
- Fitur: Interactive testing, code examples, schema export
- Keunikan: Playground dengan sample data

Minggu 3: Advanced Database & Optimization

Hari 15: Query Optimization & Select Related

Project: OptimizeMe

- Deskripsi: Query analyzer dengan N+1 detection
- Fitur: SQL query log, optimization suggestions, benchmarks
- Keunikan: Visual query plan dengan bottleneck highlighting

Hari 16: Database Indexing

Project: IndexAdvisor

- Deskripsi: Index recommendation tool based on query patterns
- Fitur: Analyze slow queries, suggest indexes, impact simulation
- Keunikan: A/B test indexes in staging environment

Hari 17: Caching Strategies

Project: CacheLayer

- Deskripsi: Multi-level caching implementation (Redis + Memcached)
- Fitur: Page cache, query cache, template fragment cache
- Keunikan: Cache warming scheduler dan invalidation patterns

Hari 18: Database Transactions

Project: PaymentGate

- Deskripsi: Payment processing dengan transaction safety
- Fitur: Atomic operations, rollback, idempotency
- Keunikan: Double-entry bookkeeping simulation

Hari 19: Raw SQL & Custom Queries

Project: ReportEngine

- Deskripsi: Business intelligence dashboard dengan complex SQL
- Fitur: Custom reports, pivot tables, data export
- Keunikan: SQL query builder dengan visual interface

Hari 20: Database Routing & Multiple DBs

Project: DataRouter

- Deskripsi: Multi-database setup dengan read replicas

- Fitur: Master-slave routing, sharding simulation
- Keunikan: Auto-failover dan connection pooling

Hari 21: Full-Text Search

Project: SearchPro

- Deskripsi: Advanced search dengan PostgreSQL full-text
- Fitur: Fuzzy search, ranking, highlighting, autocomplete
- Keunikan: Search analytics dan trending queries

Minggu 4: Testing & Debugging

Hari 22: Unit Testing

Project: TestSuite

- Deskripsi: Comprehensive test suite dengan high coverage
- Fitur: Model tests, view tests, form tests, factory fixtures
- Keunikan: Mutation testing untuk test quality

Hari 23: Integration Testing

Project: E2ETester

- Deskripsi: End-to-end testing dengan Selenium
- Fitur: User flow tests, browser automation, screenshot on fail
- Keunikan: Visual regression testing

Hari 24: Test-Driven Development

Project: TDDWorkflow

- Deskripsi: TDD practice app dengan red-green-refactor cycle
- Fitur: Test first approach, refactoring exercises
- Keunikan: Gamified TDD challenges

Hari 25: Fixtures & Factory Boy

Project: DataFactory

- Deskripsi: Test data generator dengan realistic fake data

- Fitur: Faker integration, relationship handling, seeding
- Keunikan: Snapshot testing untuk API responses

Hari 26: Django Debug Toolbar

Project: DebugDash

- Deskripsi: Performance monitoring dengan custom panels
- Fitur: SQL queries, cache hits, template rendering
- Keunikan: Historical comparison across deployments

Hari 27: Logging & Monitoring

Project: LogCenter

- Deskripsi: Centralized logging dengan ELK stack integration
- Fitur: Structured logging, error tracking, alerting
- Keunikan: Anomaly detection dalam log patterns

Hari 28: Profiling & Performance

Project: SpeedCheck

- Deskripsi: Application profiler dengan flame graphs
- Fitur: CPU profiling, memory profiling, bottleneck detection
- Keunikan: Automated performance regression tests

Hari 29-30: Capstone Project

Project: MarketPlace

- Deskripsi: E-commerce marketplace dengan seller/buyer roles
- Fitur: Product listings, cart, checkout, order management, reviews, seller dashboard, API endpoints
- Keunikan: Real-time inventory, recommendation system, multi-vendor support
- Scope: 2 hari untuk MVP dengan testing